explorations

```r
library(tidyverse)
```

```r
Sys.setenv("VROOM_CONNECTION_SIZE" = 131072 * 100)
expr_beat <- read_tsv("../../data/processed/beat_aml_expression.tsv.xz",
                      show_col_types = FALSE)

meta_beat <- read_tsv("../../data/processed/beat_aml_metadata.tsv",
                      show_col_types = FALSE)

expr_tcga <- read_tsv("../../data/processed/tcga_aml_expression.tsv.xz",
                      show_col_types = FALSE)

meta_tcga <- read_tsv("../../data/processed/tcga_aml_metadata.tsv",
                      show_col_types = FALSE)

gem <- read_tsv("../../data/processed/human_gem_gene_subsystem_summary.tsv",
                show_col_types = FALSE)
```

BEAT

```r
beat_long <- expr_beat %>%
  pivot_longer(-sample_id, names_to = "hugo", values_to = "abun") %>%
  group_by(sample_id) %>%
  mutate(rela = abun / sum(abun)) %>%
  ungroup() %>%
  mutate(hugo = if_else(hugo %in% gem$hugo, hugo, "non_metabolic")) %>%
  group_by(sample_id, hugo) %>%
  summarise(abun = sum(abun), rela = sum(rela), .groups = "drop") %>%
  left_join(gem, by = "hugo") %>%
```

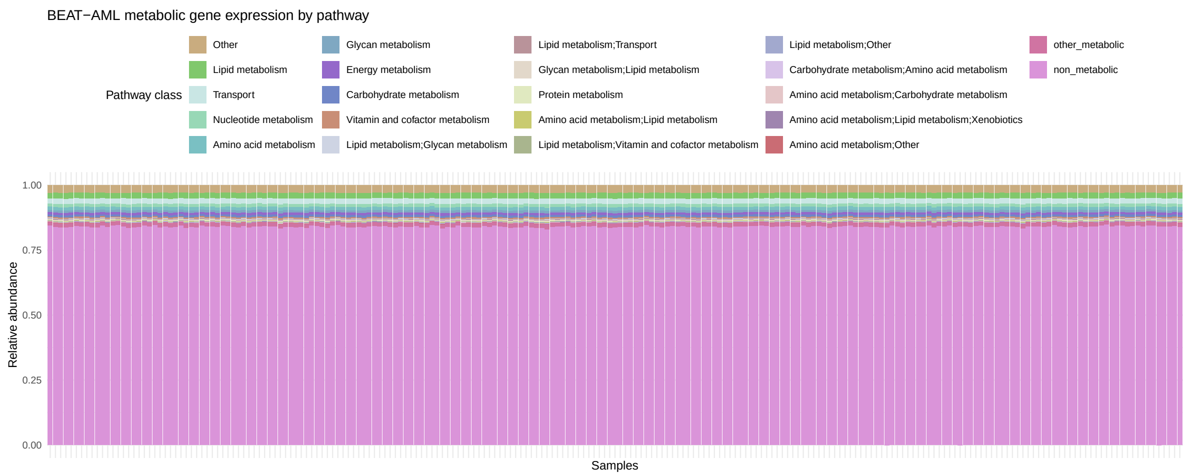
```r
  mutate(
    subsystems     = if_else(hugo == "non_metabolic", "non_metabolic", subsystems),
    pathway_class = if_else(hugo == "non_metabolic", "non_metabolic", pathway_class))
```

```r
top_pathway_class <- beat_long |>
  select(sample_id, pathway_class, abun) |>
  filter(pathway_class != "non_metabolic") |>
  group_by(sample_id, pathway_class) |>
  summarise(abun_metab = sum(abun), .groups = "drop") |>
  group_by(sample_id) |>
  mutate(rela_metab = abun_metab / sum(abun_metab)) |>
  ungroup() |>
  group_by(pathway_class) |>
  summarise(total_rela = sum(rela_metab), .groups = "drop") |>
  arrange(desc(total_rela)) %>%
  slice_max(total_rela, n = 20) %>%
  pull(pathway_class)
```

barplot

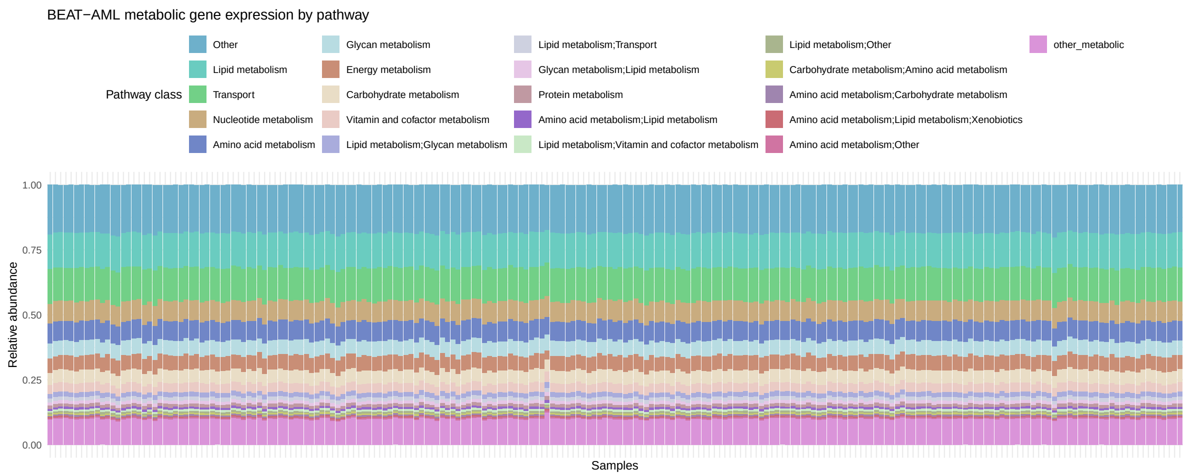
```r
pathway_order <- c(top_pathway_class, "other_metabolic", "non_metabolic")

beat_long |>
  select(sample_id, pathway_class, abun) |>
  mutate(pathway_class_fct = case_when(
    is.na(pathway_class)                  ~ NA_character_,
    pathway_class == "non_metabolic"      ~ "non_metabolic",
    pathway_class %in% top_pathway_class ~ pathway_class,
    TRUE                                  ~ "other_metabolic")) %>%
  mutate(pathway_class_fct = factor(pathway_class_fct, levels = pathway_order)) %>%
  group_by(sample_id, pathway_class_fct) |>
  summarise(abun_metab = sum(abun), .groups = "drop") |>
  group_by(sample_id) |>
  mutate(rela_metab = abun_metab / sum(abun_metab)) |>
  ungroup() |>
  ggplot(aes(x = sample_id, y = rela_metab, fill = pathway_class_fct)) +
  geom_col() +
  theme_minimal() +
  theme(axis.text.x = element_blank(),
        axis.ticks.x = element_blank()) +
```

```
  labs(x = "Samples", y = "Relative abundance",
       fill = "Pathway class", title = "BEAT-AML metabolic gene expression by pathway") +
  theme(legend.position = "top") +
  ggqualpal::scale_fill_qualpal()
```




```
beat_long |>
  select(sample_id, pathway_class, abun) |>
  filter(pathway_class != "non_metabolic") |>
  mutate(pathway_class_fct = case_when(
    is.na(pathway_class)                 ~ NA_character_,
    pathway_class %in% top_pathway_class ~ pathway_class,
    TRUE                                 ~ "other_metabolic"
  )) |>
  mutate(pathway_class_fct = factor(pathway_class_fct, levels = c(top_pathway_class, "other_
  group_by(sample_id, pathway_class_fct) |>
  summarise(abun_metab = sum(abun), .groups = "drop") |>
  group_by(sample_id) |>
  mutate(rela_metab = abun_metab / sum(abun_metab)) |>
  ungroup() |>
  ggplot(aes(x = sample_id, y = rela_metab, fill = pathway_class_fct)) +
  geom_col() +
  theme_minimal() +
  theme(axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        legend.position = "top") +
  labs(x = "Samples", y = "Relative abundance",
       fill = "Pathway class",
```

```
    title = "BEAT-AML metabolic gene expression by pathway") +
  ggqualpal::scale_fill_qualpal()
```



Dimensionality reduction

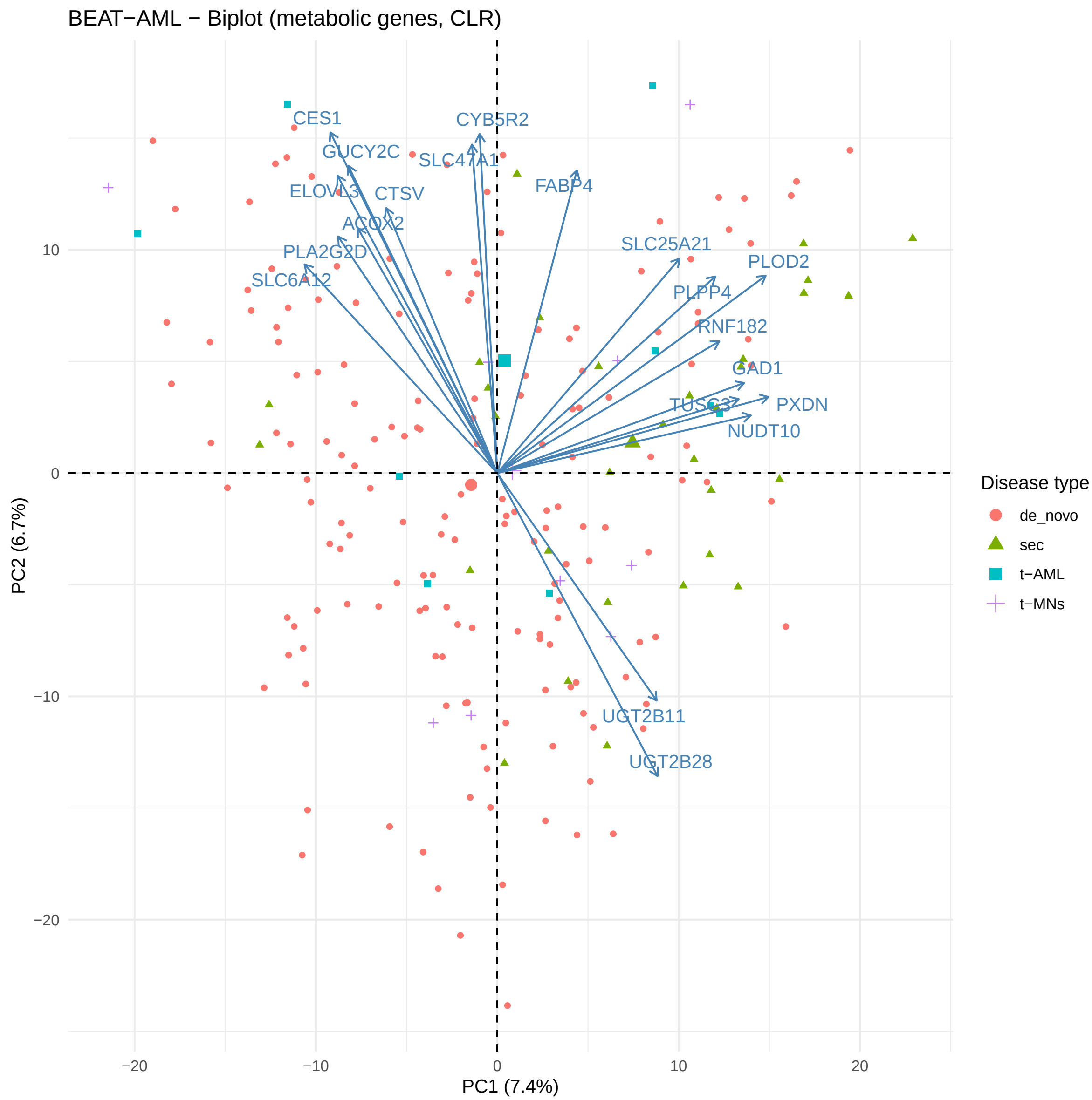
PCA

```
mat_beat <- expr_beat |>
  column_to_rownames("sample_id") |>
  as.matrix()

mat_beat <- mat_beat[, intersect(colnames(mat_beat), gem$hugo)]
mat_beat <- mat_beat[, apply(mat_beat, 2, var) > 0]

mat_beat_clr <- mat_beat |>
  zCompositions::cmultRepl(method = "CZM", z.warning = 1, z.delete = FALSE) |>
  vegan::decostand(method = "clr")
```

```
No. adjusted imputations:  62259
```

```
pca_beat <- prcomp(mat_beat_clr)
var_exp  <- pca_beat$sdev^2 / sum(pca_beat$sdev^2) * 100
```

```r
library(factoextra)

fviz_pca_biplot(pca_beat,
                geom.ind  = "point",
                col.ind   = meta_beat$Disease_Type,
                select.var = list(contrib = 20),
                repel     = TRUE,
                legend.title = "Disease type") +
  labs(x = paste0("PC1 (", round(var_exp[1], 1), "%)"),
       y = paste0("PC2 (", round(var_exp[2], 1), "%)"),
       title = "BEAT-AML - Biplot (metabolic genes, CLR)") +
  theme_minimal()
```



UMAP

```r
library(uwot)
```

```
Loading required package: Matrix
```

```
Attaching package: 'Matrix'
```

```
The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```r
set.seed(42)
umap_beat <- umap(mat_beat_clr, n_neighbors = 15, min_dist = 0.1)

tibble(sample_id = rownames(mat_beat),
       UMAP1     = umap_beat[, 1],
       UMAP2     = umap_beat[, 2]) |>
  left_join(meta_beat, by = "sample_id") |>
  ggplot(aes(x = UMAP1, y = UMAP2, color = Disease_Type)) +
  geom_point(alpha = 0.7, size = 2) +
  labs(color = "Disease type",
       title = "BEAT-AML - UMAP (all genes)") +
  theme_minimal()
```